

GPS Hijinks – Complete Solution Report

Challenge Objective

Identify the anomalous GPS log, investigate suspicious GPS data manipulation, and recover the hidden flag.

Methodology

1. Extracted gps_logs.zip and enumerated all CSV files.
2. Loaded each CSV into a dataframe for comparison.
3. Calculated point-to-point coordinate deltas to detect impossible jumps.
4. Compared flight paths, altitude trends, and speed values.
5. Investigated the largest anomaly and associated records.

Files Analysed

flight_01.csv, flight_02.csv, flight_03.csv, flight_anomaly.csv

Anomalous Flight Log

flight_anomaly.csv was identified as the manipulated dataset.

A sudden coordinate jump occurs between timestamps 290 and 300 seconds.

Before jump: Latitude=28.6202 Longitude=77.2152

After jump: Latitude=28.9500 Longitude=77.8500

Altitude simultaneously jumps to 180 m and speed spikes to 95 m/s.

Why It Is Suspicious

The drone instantly relocates tens of kilometres between two samples separated by only 10 seconds. This movement is physically impossible and is consistent with GPS spoofing or deliberate coordinate manipulation.

Recovered Flag

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FLAG{GPS_SPOOFING_DETECTED}
```

Conclusion

The manipulated flight log is flight_anomaly.csv. The GPS position jump, altitude increase, and speed spike clearly indicate tampering. The challenge flag was recovered from the anomalous data investigation.